

The Secret Life of Ink

Chromatography Mystery

Session 1: The Great Colour Race

A Mysterious Note!

Ananya found a secret note in the classroom written with a black sketch pen. She thought black was just one colour, but what if she's wrong? What if black ink is hiding a whole team of colours inside? Today, we use science to uncover the truth! [cite: 1]

Your Science Art Goals

Today, we will become "Art-Scientists" to achieve these goals:

- **Understand Mixtures:** Learn that some things are a "team" of many parts. [cite: 1]
- **Define Chromatography:** Discover the method used to separate hidden parts. [cite: 1]
- **Perform the Experiment:** Watch the "Colour Race" happen on paper. [cite: 1]
- **Become a Science Artist:** Use chromatography to create beautiful patterns. [cite: 1]

THE QUICK GUESS

What colours do you think are hidden inside a black sketch pen? Red? Blue? Yellow? Let's make a prediction before we start! [cite: 1]

What Is a Mixture?

In science, a **Mixture** is when two or more things are put together but not permanently joined. Ink is a perfect example of a hidden mixture. [cite: 1]

The Secret Team: Black ink looks like one colour, but it is actually many different colours hidden together like a team! [cite: 1]

The Fruit Chaat Analogy

Think about a bowl of **Mixed Fruit Chaat**. [cite: 1]

- It looks like one tasty snack.
- But inside, you can find pieces of Apple, Banana, and Papaya.
- Black ink is just like a "chaat" of different colours! [cite: 1]

[Visual: Comparison diagram of Mixed Fruit Chaat and a drop of Black Ink]

The Great Colour Race

If ink is a mixture, how do we separate the colours? We use a magic trick called **Chromatography**.
[cite: 1]

Definition: Chromatography is a way to separate a mixture by making its parts move at different speeds through a material. [cite: 1]

The Rainy Day Race

Imagine a race on a rainy day in your colony. [cite: 1]

- When the ground gets wet, some children run very fast.
- Some children run a bit slower on the slippery ground.
- In chromatography, **water** helps the colours "run" up the paper. Fast colours reach the top first, while slow ones stay behind! [cite: 1]

The Experiment: Uncovering the Secret

Let's set up our own Colour Race. You will be amazed at what you see! [cite: 1]

What You Need:

- A black sketch pen (water-soluble). [cite: 1]
- A strip of tissue paper or filter paper. [cite: 1]
- A small drop of water. [cite: 1]
- A flat surface (newspaper or table). [cite: 1]

ART-SCIENTIST TIP

Different brands of black pens use different colour "teams." Try two different black pens to see if the results are the same! [cite: 1]

Step-by-Step Magic

1. The Starting Line

Draw a thick black line or a big dot near the bottom of your paper strip. [cite: 1]

2. Add the "Runner"

Place a small drop of water directly on or just below the black ink. [cite: 1]

3. Watch the Race

Wait for a few minutes. Watch as the water starts to climb up the paper, carrying the ink with it! [cite: 1]

[Visual: Illustration of water carrying different colours up a tissue paper strip]

What Do You See?

As the water climbs, the "Black" ink will start to disappear. In its place, you might see: [cite: 1]

Bright Blue
Deep Purple
Warm Brown
Soft Orange

The Science Secret

The colours that you see at the **top** of the paper are the "Fast Runners." The colours that stay near the **bottom** are the "Slow Runners." [cite: 1]

Interactive: Hidden Colours Challenge

Can you guess which colours are hidden in these other sketch pens? [cite: 1]

1. THE GREEN PEN

Which two primary colours do you think are hiding in Green? [cite: 1]

Your Guess: _____ and _____

2. THE PURPLE PEN

Which two colours make up the team for Purple? [cite: 1]

Your Guess: _____ and _____

3. THE ORANGE PEN

Which team is hiding here? [cite: 1]

Your Guess: _____ and _____

Be a Science Artist!

Chromatography is not just for scientists—it's for artists too! [cite: 1]

The Colour Map Worksheet

Try chromatography with 3 different pens. In the boxes below, draw the "Colour Map" you see after the water finishes the race. [cite: 1]

[Pen 1]

[Pen 2]

[Pen 3]

Art Idea: Use these colourful strips to make a bookmark or a greeting card! [cite: 1]

Lesson Recap

You have successfully uncovered the secret life of ink! Let's remember our key findings: [cite: 1]

- **Mixtures:** Black ink is actually a "team" of many hidden colours. [cite: 1]
- **Chromatography:** We can separate these colours using water. [cite: 1]
- **The Race:** Different colours move at different speeds on paper. [cite: 1]
- **Art-Scientist:** Science can help us create beautiful things! [cite: 1]

Stay Curious!

Next time you use a sketch pen, remember the hidden team of colours inside! [cite: 1]